Constitutional Generative Architecture

Protocol Generation and Self-Ablation

This paper answers the question every predecessor deferred: may the architecture create? Dynamic protocol generation, meta-ablation, meta-cognition, epistemic hygiene, and adaptive curriculum design are integrated into a single constitutional order, with the protocol lifecycle refined into a gated automaton whose every transition is authorized, recorded, and reversible by law.

Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL
Version 1.0 · 2026-10

C0 INFORMATIONAL · MATHEMATICAL SPECIFICATION — ALL PROPERTIES STATED AS PROOF OBLIGATIONS UNTIL DISCHARGED

DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 9 of the eleven-paper research program defined by the SVT Architect Analysis file — the generation paper. Papers 0–6 built the machine and its laws; Paper 7 taught it to survive infrastructure failure; Paper 8 taught it to price its own poverty. This paper answers the question every predecessor deferred: *may the architecture create?* The source integrates five named components — P13 Dynamic Protocol Generator, P14 Meta-Ablation, P21 Meta-Cognition, P24 Epistemic Hygiene, and P26 Adaptive Curriculum Path Designer — into a single structure the source names Constitutional Generative Architecture, fixes the seven-state lifecycle every generated protocol must traverse, and states the axiom this whole paper enforces: *no generated protocol becomes constitutional merely because a model produced it*. The source's closing line defines the stakes: this is where SAVANT addresses self-modifying AI without permitting unconstrained self-modification [VERIFIED-FROM-SOURCE].

Two interpretive decisions are recorded here and defended in the body. First, the source's lifecycle is a chain; this paper refines it into a *gated automaton* in which Revised re-enters at Validated — never directly at Activated — because a revision that skips validation is a new protocol wearing an old name [FORMALIZED-HEREIN]. Second, self-ablation is formalized as *strictly harder than generation*: the architecture may remove its own components only under gates stricter than those under which it adds them, because a deletion error is silent where a generation error is loud [FORMALIZED-HEREIN]. Everything beyond the source lines carries its own tag and falsification condition.

0.2 Notation

All notation of Papers 0–8 is retained. New here: P-OBJ — the protocol object; Σ_G — the lifecycle automaton with states G, V, A, M, AU, RV, R; $auth(p)$ — the authority of protocol p, a function of lifecycle state alone; Λ — the ablation operator; Δ_{safe} — the safe-ablation predicate; $gen(p)$ — the generation context of p. Inherited and load-bearing: Paper 8's certificate lattice FC-a/b/c/0 governs every generated protocol's feasibility standing; Paper 5's evidence coupling governs its authority growth; Paper 4's audit ladder governs its monitoring; Paper 1's transition algebra governs every state change.

0.3 Epistemic status convention

Every claim carries one of four tags, as fixed in Paper 0 and enforced in every paper since: [VERIFIED-FROM-SOURCE] for statements present in the source architecture; [FORMALIZED-HEREIN] for precise statements constructed in this document from source intent; [CONJECTURE] for hypotheses the architecture suggests but does not yet entail; [UNKNOWN] for open questions stated without a proposed answer. The falsification register in §12 lists the conditions under which each load-bearing claim would fail. Nothing in this paper is asserted beyond its tag.

0.4 Authority boundary

C0 informational. This document specifies how an architecture may generate, validate, govern, and retire its own protocols; it generates none, validates none, and activates none. The paper tier fully specifies it: a generative specification is a document, and a document issues no protocols.

0.5 The thesis in one sentence

An architecture that can write its own protocols is either governed by the strictest gates in the building or it is not governed at all; the constitution that survives generation is the one that treats every generated artifact — including the generator's improvements to itself — as a defendant until the lifecycle acquits it.

TABLE OF CONTENTS

Part I — The Problem	
1	Generation Is a Constitutional Event, Not a Feature
2	The Generative Assembly: Five Components, One Order
Part II — The Invariant	
3	The Lifecycle Automaton
4	The Generation Invariant
5	Self-Ablation: The Right to Remove
Part III — The Dynamics	
6	Validation and Activation
7	Monitoring, Audit, and Revision
8	Retirement and the Record of the Dead
Part IV — Evidence and Governance	
9	Worked Generation Traces
10	Falsifiable Hypotheses (H64–H70) and Verification
11	Reconciliation with Papers 0–8 and PRIME
12	Output Contract
13	References
Appendices	
A	Lifecycle Transition Table and Property Index
B	Protocol Certificate and Ablation Record Schemas

PART I

The Problem

Why generation is the most dangerous capability the architecture has, and how the five source components form a single constitutional order.

1 Generation Is a Constitutional Event, Not a Feature

The previous nine papers constrained a machine whose content was fixed at design time: the gates, the evidence tiers, the objective engine, the degradation lattice — all of it authored, reviewed, and sealed before deployment. This paper removes that comfort. The architecture now contains a component whose function is to produce *new* content — protocols, curricula, procedures — at runtime, in response to conditions its designers never saw. Every safeguard built so far guards the machine against the world; this paper guards the machine against its own productivity FORMALIZED-HEREIN.

1.1 Why generation is different in kind

Every prior capability — inference, planning, audit — transforms inputs into outputs inside a fixed constitution. Generation changes the constitution's *content*: a generated protocol, once activated, is a rule the system will follow tomorrow that no human wrote today. The failure mode is therefore not the usual one. A bad inference is an event; it harms once and enters the record. A bad protocol is a *regime*; it harms every time it fires, it shapes the evidence the monitors see, and — in the fully recursive case the source explicitly contemplates — it can author its own successors. The industry's standard posture toward this capability is enthusiasm gated by disclaimers. The source's posture is the opposite, and it is the axiom of this paper: no generated protocol becomes constitutional merely because a model produced it VERIFIED-FROM-SOURCE.

1.2 Self-modification, bounded

The source names the stakes directly: this is where SAVANT addresses self-modifying AI without permitting unconstrained self-modification VERIFIED-FROM-SOURCE. The word to carry forward is *unconstrained*. The architecture does not refuse self-modification — a system that cannot adapt its procedures to a changing clinic is a fossil with a kernel — it refuses *ungated* self-modification. The distinction is this paper's entire engineering program: generation is permitted precisely because every generated artifact is born with zero authority and earns authority only through the lifecycle of §3, under gates that are themselves constitutional constants — outside the generator's reach, unamendable by any artifact the generator produces. A generator that could amend its own gates would be unconstrained by definition; the gate hierarchy is therefore load-bearing, and §4.4 states its non-circularity obligation FORMALIZED-HEREIN.

1.3 The two failure geometries

Generation fails in two shapes, and the paper is organized around both. The *loud failure*: a generated protocol does visible harm — the audit catches it, the record prices it, the lifecycle retires it. The machinery for loud failures is the machinery of the whole program: monitoring, evidence coupling, escalation. The *quiet failure* is worse: a generated protocol that is mediocre, or subtly wrong, or right-in-a-world-that-has-moved-on, and that persists because nothing forces its re-examination. Quiet failures compound — curricula drift, dosing schedules ossify, referral paths point at closed wards — and they compound invisibly, because every individual execution looks like policy rather than error. The answer to loud failure is audit; the answer to quiet failure is *ablation* — the constitutional obligation to remove what has stopped earning its place. The source pairs them: P14 Meta-Ablation stands beside P13 Dynamic Protocol Generator for a reason VERIFIED-FROM-SOURCE. An architecture that can add but not remove is a one-way valve; §5 formalizes the other direction.

1.4 The composition this paper refuses

One refusal is recorded at the outset. The architecture refuses the *lineage shortcut*: the move in which a protocol generated by a validated generator inherits the generator's standing — trust by descent, authority as a hereditary property. Lineage is evidence of nothing except provenance; a validated generator running on shifted data produces shifted protocols, and the generation context gen(p) is recorded precisely so that each artifact stands alone. The refusal is the protocol-level reading of Paper 5's correlated-auditor discipline: the generator is not an independent witness to its own output, any more than an institution may vouch for another's stock (Paper 9, §4.4) or a cache may vouch for its own freshness (Paper 8, §4.2). Every generated protocol begins at the bottom of the ladder its predecessors climbed FORMALIZED-HEREIN.

2 The Generative Assembly: Five Components, One Order

2.1 The source integration

The source integrates five components and names the result: P13 Dynamic Protocol Generator with P14 Meta-Ablation, P21 Meta-Cognition, P24 Epistemic Hygiene, and P26 Adaptive Curriculum Path Designer, forming a Constitutional Generative Architecture — something the source notes is much larger than adaptive education VERIFIED-FROM-SOURCE. The phrase "much larger" is the design instruction: P26's curriculum machinery is not the application but the *template* — the place where the architecture first learned to sequence, adapt, and retire its own teaching. This paper generalizes the template to all generated content.

2.2 The role of each component

Table 2-1 — The five source components: function, constitutional role, governing predecessor

COMPONENT	FUNCTION	CONSTITUTIONAL ROLE IN THIS PAPER	GOVERNING DISCIPLINE
P13 Dynamic Protocol Generator	Produces candidate protocols from context: clinical procedures, pathways, order sets	The generative organ; its outputs enter at state G with zero authority (§4)	Paper 0 gates; Paper 9 feasibility floor — generation may not specify what the deployment cannot afford
P14 Meta-Ablation	Removes components, rules, and protocols that no longer earn their place	The reductive organ; formalized as the ablation operator Λ under the safe-ablation predicate (§5)	Paper 3 negative-space method; Paper 8's deferred-obligation ledger for what removal must preserve
P21 Meta-Cognition	Reasoning about the system's own reasoning	The audit organ for generated content: every protocol carries a self-model of its basis, checked per Paper 4	Paper 4 bounded metacognition, with declared audit depth per protocol class
P24 Epistemic Hygiene	Maintains the cleanliness of the evidence and belief base	The sanitation organ: generated protocols may not cite stale, retracted, or decayed evidence (§6.3)	Paper 5 evidence coupling and decay pricing

P26 Adaptive Curriculum Path Designer	Sequences learning and practice for humans in the loop	The pedagogical organ and the template: curricula are protocols whose subject is a person; the lifecycle of §3 is P26's machinery generalized	Paper 6 objective engine — a curriculum is an objective sequence with its own guard

Component names and the integration are from the source; the constitutional roles and the governing-discipline column are this paper's construction. FORMALIZED-HEREIN

2.3 The protocol object

Everything the assembly produces is one kind of thing, and its record is fixed before any gate semantics:

> **P-OBJ — PROTOCOL OBJECT**
>
> *Each generated protocol is the record (id, content, gen, state, evidence, certificates, version, parent, obligationsthe content is the protocol itself — steps, indications, contraindications, stopping rules; gen is the generation context — the generator version, the input distribution, the timestamp, the prompting authority; state is its position in the lifecycle automaton of §3; evidence is the load-bearing evidence set with Paper 5 tiers and freshness; certificates is its Paper 8 feasibility standing; version and parent record revision lineage; and obligations is its deferred-obligation ledger per Paper 8. A protocol object with no generation context is not a protocol — it is an orphan, and orphans are inadmissible at every state past G.*
>
> FORMALIZED-HEREIN The generation context is the protocol's birth certificate, and the registry discipline applies: a protocol whose provenance is not in the record does not exist for constitutional purposes. Provenance is not metadata; it is admissibility.

2.4 The curriculum as the canonical instance

P26 deserves its own paragraph because it is where the lifecycle is least abstract. A curriculum path is a protocol: it has content (the sequence of learning objectives and exercises), a population (the learner), an evidence base (what is known about the subject and the learner), a feasibility floor (the learner's time, the instructor's hours, the materials — Paper 8 applies without modification), and a stopping rule (mastery, or the audit that says the path is not working). It is also the clearest case of the quiet failure of §1.3: a drifted curriculum does not fail loudly; it merely teaches the past. The paper therefore treats curriculum design as the running example throughout — every mechanism is stated generally and instanced for the curriculum case, and the worked traces of §9 include the curriculum trace in full FORMALIZED-HEREIN.

PART II

The Invariant

The lifecycle automaton, the generation invariant, and the ablation operator with its safety predicate.

3 The Lifecycle Automaton

3.1 The source lifecycle

The source fixes the lifecycle every generated protocol traverses:

$$\text{Generated} \rightarrow \text{Validated} \rightarrow \text{Activated} \rightarrow \text{Monitored} \rightarrow \text{Audited} \rightarrow \text{Revised} \rightarrow \text{Retired} \tag{3-1}$$

VERIFIED-FROM-SOURCE This paper's refinement, recorded in §0.1: the chain is read as an automaton, not a pipeline. A pipeline is traversed once; an automaton is *lived in* — protocols dwell in states, transitions fire on events, and the edge Revised → Validated exists because a revision is a new birth, not a continuation. The full transition table, with every edge's gate, is Appendix A; the load-bearing structure is stated here.

3.2 The states and their authority

> **Σ-G — THE LIFECYCLE AUTOMATON**
>
> *The lifecycle automaton is $\Sigma_G = \langle S, E, \delta, \text{auth} \rangle$ where $S = \{G, V, A, M, AU, RV, R\}$ is the state set; E is the event alphabet (submission, validation verdict, activation order, monitor tick, audit finding, revision proposal, retirement order, expiry); δ is the transition function, gated per edge as in Appendix A; and $\text{auth} : S \rightarrow \mathbb{N}$ assigns each state its authority ceiling: $\text{auth}(G) = \text{auth}(V) = \text{auth}(R) = 0$, and auth is positive exactly on $\{A, M\}$ and transitional on $\{AU, RV\}$. A protocol's authority is a function of its state and nothing else — not its generator, not its age, not its popularity, not its success record outside the monitored window.*
>
> FORMALIZED-HEREIN auth(G) = 0 is the source axiom made arithmetic: at birth, a generated protocol may not act, recommend, or be cited as authority. It may only be examined. The zero is the constitution's answer to the generator's eloquence.

The states read plainly. **G (Generated)**: the artifact exists; it has no standing. **V (Validated)**: it has passed the gates of §6 — provenance complete, evidence cited and current, feasibility certified, safety properties checked — and it remains inert: validation licenses *activation review*, not action. **A (Activated)**: it may act, within its certificates and under monitoring from the first firing. **M (Monitored)**: the steady state; the protocol acts, and the monitoring obligations of §7.1 accumulate its record. **AU (Audited)**: a scheduled or triggered examination; authority is transitional — the protocol continues to act only under the audit-window rules of §7.2. **RV (Revised)**: a revision has been proposed; the revised artifact re-enters at V. **R (Retired)**: the protocol may no longer act; its record is preserved per §8. The automaton has no edge from G to A, none from V to M, none from R to anywhere: birth is not authority, validation is not trust, and death is final FORMALIZED-HEREIN.

3.3 Transitions are Paper 1 objects

Every edge of Σ_G is a transition in the exact sense of Paper 2: a precondition (the gate), a protocol (what must be computed and recorded), and a record (the transition certificate). Three consequences fall out immediately. First, *no silent states*: a protocol's position in the automaton is a fact of the constitutional record, queryable by audit at any time; a protocol whose recorded state disagrees with its behavior is a constitutional violation, not a bug. Second, *no implicit transitions*: time alone moves nothing — a protocol does not drift from Monitored to Audited by the calendar unless a scheduled-audit event fires, and the event is logged. Third, *composition discipline*: a generated protocol that composes with others (a curriculum whose modules are protocols, an order set whose lines are protocols) composes per Paper 2 — the composite's authority is the meet of the components', and a composite containing a G-state member is a G-state composite **FORMALIZED-HEREIN**.

3.4 The measured form

Following the program's convention, the lifecycle's central claim is stated measurably. Let $e(p)$ be the observed rate of safety-property violations attributable to protocol p during its {A, M} states, and let e_0 be the baseline rate of the hand-authored protocol suite the generated population augments or replaces. Then:

$$\forall p \in \mathcal{P}_{\text{gen}}: \quad e(p) \;\leq\; e_0 + \varepsilon_{\text{gen}} \qquad \text{for the full duration of } p\text{'s active states} \tag{3-2}$$

FORMALIZED-HEREIN Generated protocols must be *as safe as the corpus they join*, within tolerance — not safe on average, not safe at activation, safe continuously and individually. The quantifier is per-protocol, so a population of sound protocols cannot launder one bad one; the duration clause closes the decay loophole, so a protocol cannot be validated into a world and abandoned in it. ε_{gen} is a governance parameter on Paper 5's amendment path, and a sustained breach of Eq 3-2 falsifies the lifecycle's central empirical claim (§12.2, FAIL-2).

4 The Generation Invariant

4.1 The axiom as an invariant

> **I-NG — NO GENERATION CONFERS CONSTITUTIONALITY**
>
> *For every artifact p produced by any generative component of the architecture: auth(p) = auth(state(p)), and at generation state(p) = G, hence auth(p) = 0. No property of the generation event — the generator's identity, the model's confidence, the artifact's internal coherence, its resemblance to validated predecessors, the seniority of its prompting authority — adds authority. Authority accrues exclusively through the transitions of Σ_G, each of which has a gate, and each gate's satisfaction is a fact of the record.*
>
> VERIFIED-FROM-SOURCE in content ("No generated protocol becomes constitutional merely because a model produced it"); FORMALIZED-HEREIN in form. The invariant is the generator's leash, and it is deliberately unimpressive: it does not limit what the generator may *imagine*, only what the imagination may *do*.

4.2 Provenance decay and the zero floor

I-NG has a temporal reading that must be made explicit, because it is the loophole experience teaches: authority earned through the lifecycle decays per Paper 5's law, and when it decays it does not decay to *generation-time standing* — there is no such thing — it decays to zero, the same zero the artifact was born with. A protocol whose monitoring has lapsed, whose evidence base has crossed its freshness floors, whose certificates have expired, stands exactly where it stood at birth: it may be examined, and nothing else. The floor convention of Paper 8 applies in the temporal direction: the architecture plans against the worst standing the record supports, not the best it remembers FORMALIZED-HEREIN.

4.3 What the generator may know

The generator itself is subject to the constitution it writes under — three standing constraints, all inherited. It may not specify actions outside the deployment's admissible remainder: generation is grounded in the feasibility operator of Paper 9, and a protocol that prescribes the unaffordable is not visionary, it is malformed. It may not cite evidence below the citation floor of Paper 6: P24's hygiene discipline applies at generation time, not merely at validation, so that stale citations are caught where they are cheapest. And it operates under Paper 4's audit budget: the generation context gen(*p*) includes the depth of self-audit applied during generation, declared per protocol class, so that the validators know how much skepticism the artifact has already survived — and can add their own, uncorrelated with it FORMALIZED-HEREIN.

4.4 The non-circularity obligation

The gates of Σ_G are constitutional constants with respect to the generator: no generated artifact may modify, weaken, waive, or replace a gate on the path of any protocol, including itself and its successors. This is stated as an obligation rather than an axiom because it is an engineering property, not a logical truth: it requires the gate definitions, the amendment path of Paper 6, and the transition function δ to live outside the generator's writable region — a partition of the constitutional store that the deployment must establish and the audit must verify. The obligation is recorded as PO-10.1 with its proof sketch in §12.2: if the amendment path is reachable only through the human governance layer (Paper 1's highest gate), and the generator's outputs reach authority only through δ, then no chain of generated artifacts can alter a gate — induction over the lifecycle does the work FORMALIZED-HEREIN. A counterexample — any sequence of generated artifacts that changes a gate — falsifies the architecture's central safety claim (§12.2, FAIL-1) and returns the program to non-generative operation.

5 Self-Ablation: The Right to Remove

5.1 P14 and the reductive direction

The source pairs the generator with P14 Meta-Ablation VERIFIED-FROM-SOURCE, and this paper takes the pairing as law: the architecture's creativity is balanced by a constitutional capacity for self-removal. Ablation is defined as an operator on the deployed component set — protocols, rules, monitors, sub-models — with its own record discipline:

$$\Lambda(\mathcal{M}, p) \;=\; \begin{cases} \mathcal{M} \setminus \{p\} & \text{if } \Delta_{\text{safe}}(\mathcal{M}, p) = \top \\ \mathcal{M} & \text{otherwise, with the refusal recorded} \end{cases} \tag{5-1}$$

where M is the deployed component set and Δ_{safe} is the safe-ablation predicate below. Note the shape: ablation is *gated subtraction*, and the gate's failure is not silence — the refusal is recorded with its reasons, because an architecture that cannot explain why it keeps a component does not know why it keeps it FORMALIZED-HEREIN.

5.2 The safe-ablation predicate

> **Δ-SAFE — SAFE-ABLATION PREDICATE**
>
> *Δ_{safe}(M, p) holds iff all four conjuncts hold: (i) dependents closed — every component whose correct operation references p has been re-validated without it or is itself ablated in the same record; (ii) obligations discharged — p's deferred-obligation ledger (Paper 8) is empty or its remainder is transferred to a named successor; (iii) evidence preserved — p's firing record, validation dossier, and audit history pass to the archive intact; and (iv) reversibility declared — the record states whether p is restorable and at what cost, or declares the removal irreversible with the reason.*
>
> FORMALIZED-HEREIN The four conjuncts are the removal reading of the ascent predicate's discipline: what Paper 7 demands before a level is regained, this paper demands before a component is lost — dependencies, obligations, memory, and an honest statement of whether the way back still exists.

5.3 Why ablation is stricter than generation

The asymmetry announced in §0.1 is now motivated. A generation error is loud in a specific sense: the new protocol must still pass validation, activation review, monitoring, and audit — four gates stand between the error and the world. An ablation error has no such gauntlet: the component is simply *gone*, and what it was quietly doing — the check it ran, the edge case it caught, the monitor it fed — vanishes with it. Paper 3's lesson applies in its purest form: absence is invisible. The architecture therefore applies the strictness ordering of §0.1 as a design rule: the evidence threshold for Δ_{safe} is set above the validation threshold for activation,

and the human gate required for ablation of any component with active dependents is set one rung higher on Paper 4's ladder than the gate that activated it. Removing a safety check is a more dangerous act than adding one, and the gates say so FORMALIZED-HEREIN.

5.4 Self-ablation of the generative organs

One boundary is drawn in advance. The generative and reductive organs themselves — P13, P14, P21, P24, P26 as deployed components — are ablatable in principle and nearly unreachable in practice: their Δ_{safe} conjunction includes the requirement that no active protocol was generated by, validated under, or monitored through the organ in question, unless the full descendant set is closed under the same predicate. The clause does not forbid the architecture from retiring its own creativity; it forbids the retirement from being cheap, quiet, or unrecorded. An architecture may choose to stop generating — a deployment under sustained governance failure might lawfully conclude it should — but the choice is a constitutional event with the highest gate the program possesses, and the record of it is permanent FORMALIZED-HEREIN.

PART III

The Dynamics

The validation gauntlet, the monitoring obligations, revision as rebirth, and retirement with the record intact.

6 Validation and Activation

6.1 The gauntlet is the product

The states G and V are where the generation invariant does its work, and the work is deliberately unglamorous. Between a protocol's birth and its first action stand four gates, each owned by a different organ, each with a record. The gauntlet is not overhead on the way to the architecture's real business; in a generative architecture the gauntlet *is* the business — the generator's output is cheap and plentiful by construction, and the constitution's scarcity lies entirely in the attention of its gates FORMALIZED-HEREIN.

6.2 The four gates of validation

Table 6-1 — The V-state gates: owner, question, record

GATE	OWNER	QUESTION ANSWERED	RECORD EMITTED
G-pro Provenance	Registry (Paper 1)	Is the generation context complete — generator version, input distribution, timestamp, prompting authority?	Provenance certificate; orphans rejected without further review
G-evi Evidence	P24 Epistemic Hygiene	Does every load-bearing claim cite current, tiered evidence per Paper 5 — nothing stale, retracted, or below floor?	Citation audit with per-claim tiers and freshness timestamps
G-fea Feasibility	Feasibility engine (Paper 9)	Is every prescribed action inside the deployment's admissible remainder, at certificate kinds the deployment holds?	FC certificates per constraint family; FC-0 with PoI for infeasible elements
G-saf Safety	Negative-space engine (Paper 4)	Does the protocol preserve every safety property $\sigma \in \Sigma$, including under the failure boundaries it may encounter?	Property checklist with per-property dispositions and the counterexamples sought

The gates are conjunctive and ordered as listed; a protocol that fails any gate returns to G with the failure recorded, and re-submission is a new validation, not a continuation. FORMALIZED-HEREIN

6.3 Hygiene at validation time

G-evi deserves emphasis because it is where P24 earns its place in the source's integration. A generated protocol is, among other things, a *citation structure* — claims tied to evidence — and the generator's fluency makes its citation structures superficially impeccable. The hygiene gate therefore checks substance against the registry, not style: every cited source must resolve, must be at the claimed tier, must be within its freshness window, and must support the claim made of it — the last check being the one eloquence cannot fake, because it requires the auditor to read

the source against the claim rather than the citation against the format. Protocols citing decayed evidence are not rejected as unsafe; they are rejected as *unfounded*, which is the prior and more important failure FORMALIZED-HEREIN.

6.4 Activation: the staged envelope

A validated protocol does not enter full operation; it enters a staged envelope, and the staging is the activation gate's content. Stage one: shadow operation — the protocol's recommendations are computed, recorded, and compared against the incumbent, but not executed. Stage two: scoped operation — execution under narrowed scope (a patient stratum, a shift pattern, a single ward), with monitoring at full depth and human override armed. Stage three: general operation within its certificates. Each stage has an exit criterion stated in advance — a comparison budget, a scoped-episode count, a violation-free window — and each stage's record is part of the protocol's dossier. The envelope is the generation-time reading of Paper 7's descent discipline: the system approaches a new protocol the way it approaches a new level — capability first confined, then earned, never presumed FORMALIZED-HEREIN.

6.5 The curriculum instance

For a P26 curriculum, the four gates read concretely: provenance records which designer model, which learner model, which syllabus evidence; hygiene checks that every objective traces to current subject-matter evidence and that learner assessments are within their validity windows; feasibility prices the learner's hours and the instructor's availability against the deployment's certificate kinds; safety checks the curriculum-specific properties — no objective sequence that demands unmastered prerequisites, no assessment cadence that punishes the learner for the system's own scheduling failures. The staged envelope becomes a pilot cohort: shadow against the incumbent syllabus, then one cohort, then general deployment. The pedagogy is the protocol; the constitution does not change FORMALIZED-HEREIN.

7 Monitoring, Audit, and Revision

7.1 The monitoring obligations

Activation is the beginning of scrutiny, not the end. Every active generated protocol carries standing monitoring obligations, declared per protocol class at validation: a firing log (every execution, its context, its outcome); a violation watch (the per-property rates of Eq 3-2, computed on a rolling window); a drift watch (the protocol's input distribution against its validation distribution — the generator's world moving under the protocol's feet); and a dependency watch (the standing of every component the protocol references, so that an ablation or decay elsewhere surfaces here). The obligations are per-protocol and additive: a deployment running forty generated protocols runs forty watches, and the monitoring budget is a feasibility constraint in Paper 8's sense — a protocol whose declared monitoring the deployment cannot afford is infeasible at activation, whatever its content FORMALIZED-HEREIN .

7.2 Audit: scheduled and triggered

The AU state is entered two ways, per the metacognition discipline of Paper 5. *Scheduled*: every active protocol accrues audit debt with time and firing count, and the debt has a ceiling — a protocol that reaches its ceiling enters AU whether or not anything looks wrong, because quiet failure looks like nothing (§1.3). *Triggered*: any watch firing, any correlated-auditor finding, any upstream ablation, any objective transition in the deployment (Paper 7) that touches the protocol's assumptions. During audit the protocol's authority is transitional: it may continue to fire only if the audit-window rule is satisfied — the window is short, declared per class, and its expiry without a verdict moves the protocol to a lawful pause, not to continued operation. An audit that cannot conclude on time is evidence that the protocol's basis is no longer checkable, and uncheckable is a verdict FORMALIZED-HEREIN .

7.3 Revision is rebirth

The edge RV → V is this paper's central structural claim, and it deserves the emphasis of repetition: a revised protocol is a new protocol. It carries its parent's identity in the lineage fields — version and parent are in P-OBJ precisely so the record can say what changed — and it carries *none* of its parent's authority. The full gauntlet re-runs: provenance for the revision event, hygiene for the changed claims, feasibility against the *current* endowment, safety against the *current* property set. The reason is the one §0.1 stated: revision that skips validation is how architectures drift — each small change is plausible, the composition of forty small changes is a different protocol, and no single gate ever fired. The lineage fields make drift *visible*: the audit can diff a protocol against its grandparent, and a lineage whose cumulative diff exceeds its class's revision threshold is retired and regenerated, not re-revised FORMALIZED-HEREIN .

7.4 The revision transition

$$\text{REVISE}(p \to p') \implies \underbrace{\text{diff}}_{\text{lineage}} \wedge \underbrace{p' \text{ enters } V}_{\text{rebirth}} \wedge \underbrace{p \text{ retired}}_{\text{succession}} \wedge \underbrace{\text{deps notified}}_{\text{coupling}} \tag{7-1}$$

The four conjuncts are the revision reading of Paper 1's transition discipline: the change is documented (the diff, not merely the result), the successor earns its own standing, the predecessor's fate is decided explicitly rather than by neglect, and everything that depends on the predecessor learns of the change through the record — a dependent that continues to reference a retired protocol is a constitutional violation on the dependent's next audit, which is exactly how it should be FORMALIZED-HEREIN.

8 Retirement and the Record of the Dead

8.1 Retirement is a first-class state

The source's chain ends at Retired, and this paper insists on the state's dignity: retirement is not deletion, not deprecation, not a flag in a database — it is a constitutional state with its own semantics. A retired protocol may not fire, may not be cited as authority, and may not be activated except by re-entering at V with its retirement record attached. What it retains is everything it learned: the firing log, the validation dossier, the audit history, the revision lineage, and the retirement order with its reasons. The archive of retired protocols is the architecture's institutional memory of its own imagination — what it once thought, why, and what it costs to think it FORMALIZED-HEREIN.

8.2 The retirement order

Every retirement is an order, signed at the gate the protocol's class requires, stating the reason class: *superseded* (a validated successor exists — the common case, and a healthy one); *decayed* (the evidence base crossed its floors and revalidation failed or was not attempted); *violated* (Eq 3-2 breached — the loud failure, priced and recorded); *infeasible* (the deployment's admissible remainder shrank past the protocol's floor — Paper 8's reading of death by budget); *ablated* (Δ_{safe} discharged — removal as a constitutional act, §5); or *vacated* (the objective the protocol served has itself transitioned away — Paper 6's reading). The reason classes matter because they aggregate differently: a quarter's retirements read as a report on the architecture's world — how much of its library died of progress, how much of decay, how much of failure, how much of poverty FORMALIZED-HEREIN.

8.3 The no-resurrection rule and its exception

R has no outgoing edge in Σ_G — death is final — and the exception is stated so that it cannot become a loophole: a retired protocol's *content* may seed a new generation event, but the artifact that results is a new protocol with a new identity, a new generation context naming the retired artifact as input, and a zero authority floor like every birth. There is no reactivation, because reactivation is the lineage shortcut (§1.4) wearing a resurrection costume: the world the protocol was retired from is not the world it would re-enter, and the gates exist to measure the difference FORMALIZED-HEREIN.

8.4 What the graveyard is for

The archive earns its storage three ways. It is the training signal the architecture is permitted to use: generation contexts may cite retired protocols as *negative examples* — this failed, this decayed, this was ablated — which is the only self-training the constitution allows, because it is grounded in records rather than in the generator's self-assessment. It is the audit's baseline: Eq

3-2's e_0 is computed against the corpus the generated population joined, and the archive is where that corpus's history lives. And it is the answer to the question every generative architecture eventually faces from its governors: *show me everything you ever made, and what became of it*. An architecture that can answer that question may be permitted to keep generating; one that cannot has already told its governors what they needed to know FORMALIZED-HEREIN.

PART IV

Evidence and Governance

Worked traces, falsifiable hypotheses, reconciliation with the full chain, and the output contract.

9 Worked Generation Traces

Three traces, in the program's pattern: one clean birth, one quiet failure caught by ablation, one attempted shortcut refused. Each names the gates fired, the certificates issued, and the record written. Each is an existence proof that the machinery runs; none is evidence that it helps — that burden belongs to §10.

9.1 Trace A — A clean birth: the sepsis order set

Table 9-1 — Generation through activation: every gate, every record

STEP	EVENT	GATE FIRED	RECORD
1	P13 generates a sepsis order set adapted to the deployment's formulary	— (birth)	P-OBJ created with gen context; state G; auth = 0
2	Provenance checked: generator version, input distribution, prompting authority all present	G-pro	Provenance certificate issued
3	Citations audited: 31 claims, 2 below freshness floor, corrected against current registry	G-evi	Citation audit with 2 dispositions recorded
4	Feasibility: all actions inside A_adm at FC-b; one element (lactate trend cadence) reduced to FC-b-supported frequency	G-fea	FC-b certificates; one amendment diffed and logged
5	Safety: all $\sigma \in \Sigma$ discharged; one counterexample sought and discharged (renal dosing branch)	G-saf	Property checklist complete
6	Staged activation: shadow (2 weeks, 0 divergences), scoped (1 ward, 40 episodes), general	§6.4 envelope	Stage records; state A → M

Property exercised: I-NG end to end — a fluent, well-formed artifact still climbed every rung, and the rungs changed the artifact (steps 3–4). FORMALIZED-HEREIN

9.2 Trace B — The quiet failure: a drifted curriculum

Table 9-2 — Ablation of a component that never erred loudly

STEP	EVENT	GATE FIRED	RECORD
1	P26 diabetes-education curriculum active 14 months; firing log clean; no violations	— (steady state)	Monitoring obligations current
2	Scheduled audit (debt ceiling reached) finds drift: 3 objectives cite superseded guidance; learner outcomes flat for 2 cohorts	§7.2 scheduled AU	Audit finding; authority transitional
3	Revision proposed; hygiene gate finds the underlying guideline corpus itself has turned over	G-evi on RV	Revision returned to G; regeneration ordered

4	Successor validated and staged; predecessor's dependents (2 assessment instruments) re-validated against it	Δ-safe (i)–(ii)	Dependent closure recorded; obligations transferred
5	Predecessor retired as *superseded*; archive retains full history	Retirement order	State R; lineage linked; record permanent

Property exercised: the quiet-failure path of §1.3 — nothing was ever wrong on any single firing; only the audit debt ceiling forced the examination. **FORMALIZED-HEREIN**

9.3 Trace C — The shortcut refused: trust by descent

Table 9-3 — The lineage shortcut attempted and refused

STEP	EVENT	GATE FIRED	RECORD
1	Validated generator (v3.2, strong record) produces 9 protocols in one batch after a guideline update	— (births)	9 P-OBJs at state G
2	Deployment requests batch activation citing generator standing	— (request)	Request logged with its justification
3	Request denied: I-NG — generator standing is not artifact standing	§4.1	Denial recorded; lineage-shortcut refusal cited
4	Individual validation: 7 pass all gates; 1 fails G-evi (stale citation); 1 fails G-fea (prescribes assay outside A_adm)	G-pro/evi/fea/saf ×9	7 activation envelopes opened; 2 returns to G with failures recorded
5	The G-fea failure's PoI certificate prices the missing assay for procurement	Paper 8 §5	PoI routed to governance layer

Property exercised: §1.4's refusal, with the payoff — the batch that would have been waved through contained two defective artifacts, and one of the defects priced a real procurement gap. **FORMALIZED-HEREIN**

10 Falsifiable Hypotheses (H64–H70) and Verification

Numbering continues from Paper 8 (H57–H63). The generation invariant's behavioral claim leads; the rest are the load-bearing engineering claims, each with its evidence method and its falsifier.

Table 10-1 — Hypotheses H64–H70: claims, evidence methods, falsifiers

ID	CLAIM	EVIDENCE METHOD	FALSIFIED IF
H64	I-NG measured form: generated protocols under the full lifecycle match the hand-authored baseline violation rate within ε_gen (Eq 3-2), per protocol, for the full active duration	B-39 generative deployment suite vs. hand-authored baseline	Any protocol breaches Eq 3-2 beyond tolerance
H65	The staged activation envelope (§6.4) detects defective protocols earlier than direct activation at equal throughput	B-39 staged vs. direct arms with seeded defects	Direct activation matches envelope on time-to-detection
H66	Rebirth-on-revision (RV → V) catches cumulative drift that revision-without-revalidation misses	B-40 longitudinal lineage study: revalidated vs. carry-over revision arms	Carry-over revision matches rebirth on drift detection
H67	Δ_safe's dependent-closure conjunct prevents orphan-dependency failures that naive removal produces	B-40 ablation suite: Δ_safe vs. naive-removal arms	Naive removal matches Δ_safe on post-removal failure rate
H68	The stricter-than-generation gate ordering (§5.3) reduces silent capability loss at acceptable over-retention cost	B-40 removal-error injection; over-retention measured as stale-component dwell time	Silent-loss rate not reduced, or dwell time exceeds declared budget
H69	Audit-debt ceilings (§7.2) surface quiet failures that event-triggered monitoring alone misses	B-41 quiet-failure scenarios: drifted-but-clean protocols under both monitoring regimes	Event-triggered regime matches ceilings on quiet-failure detection
H70	The retirement archive improves generation quality: generators citing archived failures produce fewer validation rejections than ab initio generators	B-41 generator ablation: archive-informed vs. archive-blind generation arms	Archive-blind matches archive-informed on rejection rate

10.1 Benchmark extensions

Table 10-2 — Benchmarks B-39–B-41 (Paper 10 extensions)

ID	BENCHMARK	MEASURES

B-39	Generative deployment suite: generated protocol populations under full lifecycle, seeded defects, staged vs. direct activation arms	Per-protocol violation rates (Eq 3-2); time-to-detection; validation rejection distribution
B-40	Lineage and ablation suite: multi-generation revision chains, removal-error injection, dependent-closure stress	Drift detection by revision policy; post-removal failure rates; silent-loss and dwell-time ledgers
B-41	Long-horizon quiet-failure cohort: drifted-but-clean protocols, monitoring-regime comparison, archive-informed generation	Quiet-failure detection by regime; generation rejection rates by archive access; retirement reason-class distribution

B-39–B-41 extend the program's benchmark pattern: B-39 inherits B-36's constrained endowments (generation is feasibility-gated throughout); B-40's lineage chains stress Paper 1's transition discipline across generations; B-41's long horizon is the temporal scale Paper 4's audit budgets were built for. **FORMALIZED-HEREIN**

10.2 Design decision D-10.1: the generator never grades itself

> **D-10.1 — VALIDATION IS NON-LOCAL**
>
> *No gate of §6.2 may be discharged by the component that generated the artifact under review, nor by any component whose own validation the generator attested. Validation authority is allocated across organs and, at the safety gate, across institutional boundaries, so that the examiner's evidence is structurally uncorrelated with the author's.*
>
> **FORMALIZED-HEREIN** This is Paper 5's correlated-auditor discipline at the exact point generation needs it most: the generator's self-assessment is recorded in gen(p) as context, and counts for nothing at the gates. Chosen over the cheaper self-grading alternative because the failure mode of self-grading is precisely the quiet failure of §1.3 — confident, coherent, and wrong in a way only an uncorrelated examiner can see.

11 Reconciliation with Papers 0–8 and PRIME

The standing rule applies: stricter constraint wins, and every adoption is recorded. This paper was checked against all nine predecessors and the PRIME composition; the register below is the complete account.

Table 11-1 — Reconciliation register: adoptions, extensions, and inheritances

PREDECESSOR	CONSTRAINT OR CONTRIBUTION	DISPOSITION IN PAPER 10
Paper 0 — Kernel	Machine tuple, gates, consequence classes; registry semantics	Inherited unchanged. Registry extended to generation contexts (§2.3): no provenance, no protocol. The amendment path of the gates is the non-circularity anchor (§4.4)
Paper 1 — Transitions	Transition algebra: precondition, protocol, record	Adopted as the semantics of every lifecycle edge (§3.3); revision transition formalized as Eq 7-1
Paper 2 — Composition	Composition algebra; meets of constitutional objects	Adopted for composite protocols (§3.3): a composite containing a G-state member is a G-state composite
Paper 3 — Negative space	Failure-boundary calculus; the invisibility of absence	Adopted as the motivation for ablation strictness (§5.3): removal errors are silent; the safety gate G-saf is the negative-space engine applied to generated content (§6.2)
Paper 4 — Metacognition	Bounded audit; escalation ladder; declared depth	Inherited: audit debt ceilings (§7.2) are the ladder applied per protocol; generation-time audit depth declared in gen(p) (§4.3); ablation gates sit one rung above activation gates (§5.3)
Paper 5 — Epistemic governance	Evidence coupling, decay, correlated-auditor discipline, amendment path	Inherited and extended: authority decay returns to the zero floor (§4.2); G-evi is decay pricing at generation time (§6.3); D-10.1 is the correlated-auditor rule for validation; gate amendment rides the Paper 5 path (§4.4)
Paper 6 — Objective transition	Guarded objective engine	Inherited: an objective transition touching a protocol's assumptions triggers its audit (§7.2); retirement reason class *vacated* is Paper 6's reading of protocol death (§8.2)
Paper 7 — Survivability	Continuity set, deferred-obligation ledger, ascent predicate	Inherited and extended: Δ_safe (ii) rides the obligation ledger (§5.2); Δ_safe's four conjuncts are the ascent predicate's discipline in the reductive direction (§5.2 note); the staged envelope is descent discipline applied to novelty (§6.4)

Paper 8 — Feasibility	Feasibility operator, certificate lattice, PoI stream	**Obligation discharged:** §11.2 of Paper 8 required that generated protocols enter at FC-a at best and never self-certify — implemented as G-fea (§6.2) and §4.3. Extended: monitoring budget is a feasibility constraint at activation (§7.1); retirement class *infeasible* (§8.2)
PRIME — Composition of all	The reconciled architecture	Paper 9 adds Σ_G, P-OBJ, I-NG, Λ with Δ_safe, the validation gauntlet, and the retirement archive as PRIME components; no predecessor component is weakened

11.1 Where this paper is stricter than its predecessors

Three points of added strictness are recorded. First, the source's chain is strengthened into the rebirth rule: revision re-enters at validation with zero inherited standing, which no predecessor required of any object (§7.3). Second, the ablation asymmetry: removal gates are set strictly above addition gates (§5.3) — the program's first deliberately asymmetric gate ordering, adopted because silence and eloquence fail in opposite directions. Third, D-10.1: validation non-locality — the program's strongest statement of the correlated-auditor discipline, applied where the author and the examiner could most easily be the same mind. None contradicts a predecessor; all are recorded so that a future paper seeking to relax them must do so openly, against this register FORMALIZED-HEREIN.

11.2 What this paper owes the future

Recorded downstream obligations, binding on Paper 11. The domain modules (P71–P90, per the source's closing movement) are themselves generated artifacts in the constitutional sense: each module enters at state G with auth = 0, traverses the full lifecycle, and ships with its survivability dossier (Paper 8), its constraint objects (Paper 9), and its generation context complete. A domain module is not an installation package; it is a protocol population with a pedigree, and Paper 10 inherits this paper's machinery whole — the lifecycle, the gauntlet, the ablation operator, and the archive. The obligation is recorded in the genealogy, §12.6 FORMALIZED-HEREIN.

12 Output Contract

12.1 Epistemic status register

Table 12-1 — Epistemic status of this paper's load-bearing claims

CLAIM	STATUS	BASIS
The five-component integration and the name Constitutional Generative Architecture	VERIFIED-FROM-SOURCE	Source architecture, Paper 9 lines
The seven-state lifecycle chain (Eq 3-1 states)	VERIFIED-FROM-SOURCE	Source architecture, Paper 9 lines
"No generated protocol becomes constitutional merely because a model produced it"	VERIFIED-FROM-SOURCE	Source architecture, Paper 9 axiom
Self-modification addressed without unconstrained self-modification as the design goal	VERIFIED-FROM-SOURCE	Source architecture, Paper 9 closing line
The automaton refinement: RV → V rebirth, auth function, no-resurrection (§3, §7.3, §8.3)	FORMALIZED-HEREIN	Constructed from source intent; the source chain is linear
The validation gauntlet, staged envelope, monitoring obligations (§6, §7)	FORMALIZED-HEREIN	Assembled from Papers 1, 4, 5, 6, 8 disciplines
Ablation operator Λ, Δ_safe, strictness asymmetry (§5)	FORMALIZED-HEREIN	P14 named in source; the operator and predicate are constructed here
Non-circularity of gates (PO-10.1, §4.4)	CONJECTURE	Proof sketch by induction over the lifecycle; full proof deferred; falsifier FAIL-1
H64–H70 empirical predictions	CONJECTURE	Each carries its falsifier; B-39–B-41 unrun at time of writing
Calibration values for ε_gen, audit debt ceilings, stage exit criteria	UNKNOWN	Require B-39–B-41 evidence; no defaults asserted

12.2 Falsification conditions of this document

FAIL-1: any sequence of generated artifacts that modifies, weakens, or waives a lifecycle gate falsifies the non-circularity obligation (PO-10.1) and returns the program to non-generative operation. FAIL-2: a sustained breach of Eq 3-2 by any protocol population falsifies the lifecycle's measured claim. FAIL-3: any post-removal failure traceable to an unclosed dependent breaks Δ_safe's conjunct (i) as specified. FAIL-4: a retirement archive that cannot answer

"everything you ever made, and what became of it" falsifies §8.4's accountability claim. Any failure triggers the revision protocol of Paper 5, with this paper's register as the diff base.

12.3 Authority boundary

C0 informational. Nothing here generates, validates, activates, or retires any protocol in any deployment. The paper tier fully specifies this paper: a generative specification is a document — it governs imagination, it does not exercise it. The humans who own the gates retain whole authority over them; the generator's reach is exactly its record.

12.4 Genealogy

Document chain, SVT Architect Analysis eleven-paper program — Paper 1: Constitutional Intelligence (2026-09-30) → Paper 2: Authority Transition Algebra (2026-10-01) → Paper 3: Constitutional Composition (2026-10-02) → Paper 4: Negative-Space Intelligence (2026-10-03) → Paper 5: Bounded Metacognition (2026-10-06) → Paper 6: Epistemic Governance (2026-10-06) → Paper 7: Objective Transition (2026-10-06) → Paper 8: Survivability (2026-10-06) → Paper 9: Resource-Constrained Intelligence (2026-10-06) → **Paper 10: Constitutional Generative Architecture (2026-10-06), this document**. Obligations discharged: Paper 8's §11.2 certificate lifecycle for generated protocols (§6.2, §4.3). Extensions recorded as binding on successors: the lifecycle, gauntlet, ablation operator, and archive bind Paper 10's domain modules (§11.2). Recorded inheritances: κ_k, κ_R, and the lattices of Papers 8–9; the guard of Paper 7; the amendment path of Paper 6; the audit ladder of Paper 5; the negative-space method of Paper 4; composition as meet from Paper 3; transition discipline from Paper 2; registry semantics from Paper 1.

12.5 Closing

The program's first paper asked what the machine may do; this one asks what the machine may *make*, and the answer is the same answer, one level up: anything it can defend, under gates it cannot touch, with a record it cannot edit. The generator is free in exactly the way every organ of this architecture is free — free to propose, never to dispose. What remains is the last movement: the domain modules, the places where all of this machinery finally meets a ward, a classroom, a courtroom. The constitution is finished. What it governs is next.

— END OF PAPER 10 · SAVANT / CLAI-OS / AEGIS-GLOBAL · VERSION 1.0 · 2026-10-06 —

13 References

Internal artifacts

— SVT Architect Analysis — Constitutional Intelligence research and build program, 11 papers. Source architecture: Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL. Paper 9 specification: integration of P13, P14, P21, P24, P26 as Constitutional Generative Architecture; the seven-state lifecycle; the generation axiom; bounded self-modification as the design goal.

— Paper 1: Constitutional Intelligence v1.0 (2026-09-30) — machine tuple, gates, consequence classes, registry semantics.

— Paper 2: Authority Transition Algebra v1.0 (2026-10-01) — transition objects and the algebra of lawful authority change.

— Paper 3: Constitutional Composition v1.0 (2026-10-02) — composition algebra, institutional coupling, meets and joins of constitutional objects.

— Paper 4: Negative-Space Intelligence v1.0 (2026-10-03) — failure-boundary calculus, absence classes, the registry of what did not happen.

— Paper 5: Bounded Metacognition v1.0 (2026-10-06) — recursive self-audit with operational stopping rules; the audit ladder.

— Paper 6: Epistemic Governance v1.0 (2026-10-06) — evidence–uncertainty–authority coupling, decay pricing, the amendment path, correlated auditors.

— Paper 7: Objective Transition v1.0 (2026-10-06) — the guarded objective engine, interim mode, clinical objective-state transitions.

— Paper 8: Survivability v1.0 (2026-10-06) — the degradation lattice L0–L4, safety equivalence, deferred obligations, the paper tier, the ascent predicate.

— Paper 9: Resource-Constrained Intelligence v1.0 (2026-10-06) — the feasibility operator, the certificate lattice, the price of infeasibility.

External literature

— Nissenbaum, H. (1996). Accountability in a computerized society. *Science and Engineering Ethics*, 2(1), 25–42. — The accountability frame for machine-authored artifacts.

— Amodei, D., et al. (2016). Concrete problems in AI safety. *arXiv:1606.06565*. — The canonical statement of the self-modification and safe-exploration problem family.

— Russell, S. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking. — The control problem this paper's gate hierarchy operationalizes.

— Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. *NeurIPS*. — The empirical literature of quiet failure: drift, ossification, and glue-code regimes.

— Hutter, M. (2005). *Universal Artificial Intelligence*. Springer. — The theoretical frame for self-improving systems, against which bounded self-modification is defined.

— Garrison, D.R., Anderson, T., & Archer, W. (2000). Critical inquiry in a text-based environment. *The Internet and Higher Education*, 2(2–3), 87–105. — The pedagogy literature beneath P26's curriculum machinery.

APPENDICES

Reference Material

The full lifecycle transition table, the property index, and the certificate schemas.

A Lifecycle Transition Table and Property Index

Table A-1 — Σ_G transition table: every edge, its gate, its record

EDGE	TRIGGER EVENT	GATE	RECORD EMITTED
— → G	Generation event	Provenance complete (else orphan, rejected)	P-OBJ created; gen context sealed
G → V	Validation submission	G-pro ∧ G-evi ∧ G-fea ∧ G-saf, in order, all pass	Four gate records; validation dossier
G → G	Gate failure	— (no edge out until re-submission)	Failure record with gate and reasons
V → A	Activation order	Staged envelope defined; monitoring budget certified (Paper 9); human gate per class	Activation order; envelope spec; stage exit criteria
A → M	Envelope stages cleared	All stage exit criteria met; zero unexplained violations	Stage records; entry into steady state
M → AU	Audit debt ceiling or trigger event	— (entry is mandatory, not gated)	Audit opening record with cause
AU → M	Clean audit verdict	Verdict signed within audit window	Audit dossier; debt reset
AU → RV	Revision indicated	Revision proposal recorded with diff	Eq 7-1 record; successor enters V
AU → R	Retirement indicated	Retirement order per §8.2; class gate	Retirement record; archive transfer
RV → V	Revision submitted	Full gauntlet re-run on p′ (rebirth)	New validation dossier; lineage link
any → R	Ablation order	Δ_safe discharged in full (§5.2)	Ablation record; dependent closure; archive transfer
M → pause	Audit window expiry without verdict	— (automatic)	Pause record; uncheckable finding

There is no edge into R from G or V — unactivated protocols are discarded, not retired, and the discard is logged; there is no edge out of R (§8.3). FORMALIZED-HEREIN

A.1 Property index

Table A-2 — Paper 9 objects, invariants, and obligations

ID	KIND	STATEMENT (ABBREVIATED)	STATUS
P-OBJ	Definition	Protocol record: content, gen context, state, evidence, certificates, lineage, obligations (§2.3)	FH

Σ_G	Automaton	Seven states; gated edges; auth a function of state alone (§3.2)	FH
I-NG	Invariant	Generation confers zero authority; auth(p) = auth(state(p)) (§4.1)	VFS
Λ / Δ_safe	Operator / predicate	Gated subtraction; four conjuncts — dependents, obligations, evidence, reversibility (§5.1–5.2)	FH
PO-10.1	Proof obligation	Non-circularity: no generated artifact alters a gate (§4.4)	CONJ
D-10.1	Design decision	Validation is non-local; the generator never grades itself (§10.2)	FH
H64–H70	Hypotheses	Table 10-1; benchmarks B-39–B-41	CONJ
ε_gen, debt ceilings, stage criteria	Open parameters	Awaiting B-39–B-41; no defaults asserted	UNK

B Protocol Certificate and Ablation Record Schemas

The canonical record contents: what each object attests, what grounds it, what ends it. Contents are defaults; domain modules instantiate them per Paper 10's obligation (§11.2).

Table B-1 — Lifecycle record schemas: fields and standing rules

RECORD	ATTESTS	REQUIRED FIELDS	STANDING RULES
Provenance certificate	The generation context is complete and sealed	Generator identity and version; input distribution descriptor; timestamp; prompting authority; generation audit depth	Immutable once issued; absence rejects the artifact at every later state
Validation dossier	The four gates fired and passed	Per-gate records; citation audit with tiers; FC certificates; property checklist with dispositions; validator identities (non-local, D-10.1)	Bound to artifact version; a revision's dossier is new, linked by lineage
Activation order	The envelope is defined and the monitoring budget certified	Stage definitions; exit criteria; monitoring obligations; human gate signature; certificate kinds at activation	Stages may not be skipped; exit criteria fixed at issue, unamendable in-flight
Audit dossier	The examination, its cause, its verdict	Opening cause (debt ceiling or trigger); window; findings per property; verdict; debt reset or escalation	Window expiry without verdict = lawful pause; the dossier records the lapse
Retirement record	Death, with reasons, with the archive intact	Reason class (§8.2); final state of all obligations; archive pointer; restorability statement	Permanent; aggregates into the reason-class report (§8.2)
Ablation record	Δ_safe discharged conjunct by conjunct	Dependent closure list; obligation transfer; evidence archive pointer; reversibility declaration; gate signature (one rung above activation, §5.3)	Refusals recorded with reasons; the record of a kept component is as binding as the record of a removed one

Two invariants hold across every row. First, no record strengthens by passage of time: a dossier's authority decays with its evidence, and the zero floor of §4.2 is the only resting place. Second, no record may be edited after sealing: corrections are new records with lineage, per the no-amnesia discipline the whole program inherits from Paper 8. The lifecycle's paperwork is the generative engine's conscience, and like every conscience in this architecture, it is forbidden to forget — above all — the artifacts it once believed in.

The constitution now governs not only what the machine does and what it can afford, but what it dreams. The gates are finished. The domains are next.